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Hardware · Software · UX · IoT · R&D

WORK EXPERIENCE

schwaaaa

09/2024 – Present

Co-Founder / Principal Solutions Architect & Researcher

AI R&D Solutions Architect Data Engineering Video Processing Technical Documentation

AI / Claude / Python / Rust / Media Processing / Automation / OpenClaw / Blockchain / Linux

- Conduct ongoing R&D across cloud and local LLMs, spanning data engineering, model integration, and custom hardware/software systems.
- Built video processing tools using FFmpeg and OBS integrations, alongside web automation platforms for content creators.
- Developed software for archiving physical media (e.g. VHS, DVD) for AI training pipelines using specialized hardware capture workflows.
- Archived software libraries and built decentralized WebGL solutions using the Bitcoin Ordinals protocol.
- Provide Solutions Architecture consulting for early-stage MVPs and rapid prototypes across hardware and software.

Motional

04/2023 – 07/2024

Principal UX Hardware/Software Engineer

Autonomous Vehicles Solutions Architect User Testing Cybersecurity Full Stack Application Design

Arduino / Raspberry Pi / Node.js / Javascript / Python / Cellular / AWS / Automation / Linux / IoT / DevOps

- Product technical advisor focused on cross-team communication between Engineering, Operations, and Product.
- Led a small team creating CES-quality user journey prototypes for VIP, C-Suite, and dignitary showcases.
- Built IoT prototypes for the Motional/Uber delivery collaboration for the Product team.
- Managed a VR team delivering User Journey development, HMI experiences, and patent-awarded robotics use cases.
- Served as Product team liaison for Cybersecurity, validating Product tests against updated protocols.

Motional

04/2021 – 04/2023

Senior UX Hardware/Software Engineer

Autonomous Vehicles Solutions Architect Procurement Full Stack Application Design

Arduino / Raspberry Pi / Node.js / Javascript / Python / Cellular / AWS / Automation / Linux / IoT / DevOps

- Led technical development, cross-team coordination, and material sourcing for Hyundai Ioniq 5 EV prototype vehicles.
- Created a modular software/hardware platform enabling Product teams to validate Engineering use cases in real-world settings.
- Hardware/software integration lead for UI/UX accessibility testing with Lyft, C-Suite, Hyundai, Kia, and various dignitaries.

Motional

04/2020 – 04/2021

UX Hardware/Software Engineer

Solutions Architect User Testing QA Full Stack Application Design

Arduino / Raspberry Pi / Node.js / Javascript / Python / Cellular / AWS / Automation / Linux / IoT / DevOps

- Created an IoT vehicle hardware/software sandbox for Product testing a ride-share network with Cloud and Engineering teams.
- Validated automotive and experimental hardware components against Design & Research team use cases.
- Built a VR platform for validating User Journey experiences and informing internal HMI development in Engineering.

Twilio

08/2018 – 07/2019

Developer Evangelist

Cellular IoT Community Outreach Technical Documentation

Arduino / Raspberry Pi / Node.js / Go / Javascript / React / AWS / Azure / Linux / Git

- Inspired and equipped Twilio IoT developers through blog content, tutorials, and quick-start guides using cellular hardware.
- Sponsored hackathons, hosted IoT workshops, and participated in panels on the cellular IoT development process.
- Provided specialized IoT solutions for developers using the T-Mobile Narrowband kit and Twilio SIM.
- Designed the SIGNAL Smart Door — an interactive Narrowband product demo for Twilio SIGNAL 2018.
- Created the TwilioIoT GitHub repository for sharing IoT code and snippets internally and externally.

yack.com

09/2017 – 06/2018

Full Stack Developer

Web Chat SMS Cloud

Javascript / Node.js / Express / React / AWS EC2 / DynamoDB / WebSockets

- Designed and developed a text-based notification system for Emergency Services over web chat using Twilio.
- Built the back-end in Node.js and architected cloud infrastructure from scratch using AWS EC2, S3, and DynamoDB.
- Created an API for receiving SMS from clients and updating user accounts via WebSockets.
- Installed a prototype user base using Auth0, created profile templates, and added an admin log viewer for saved calls.

Multimedia Innovations

01/2017 – 08/2017

Full Stack Developer

Text-to-Speech Automation Transcription

Javascript / Node.js / Express / IBM Watson / YouTube API

- Invented an automation method that takes a YouTube link and generates a searchable transcription, audio, and video webpage.
- Created features including keyword highlighting, quote/word save & share, and keyword-based playback.

The Semisphere Group

07/2016 – 01/2018

Lead R&D Hardware & Software Developer

VR / AR HCI Neuroscience Research

Raspberry Pi / Python / Linux / MaxMSP / Jitter / Unreal Engine / Arduino / Android / Processing

- Designed hardware/software solutions for Rhode Island Space Grant Consortium and Brown University's Planetary Geology dept. under Seth Horowitz, PhD & Peter H. Schultz, PhD.
- Developed Oream — a modular hardware/software anti-nausea solution to reduce the effects of vertigo.
- Built custom VR software for navigating real-time 3D geological models with joystick control for Brown University.
- Created custom prototyping software to test prosthetic hand gestures.

The Engine Institute

01/2016 – 05/2016

Lead R&D Hardware & Software Developer

Brainwave Algorithms LED Sculpture HCI

MaxMSP / Jitter / MindWave / Emotiv / Arduino / Neurosky

- Developer of MindDraw — a real-time interactive brainwave visualizer driven by Meditation and Attention patterns via the NeuroSky MindWave.
- Created software to capture and filter user brainwave patterns for developing experience presets.
- Designed a physics engine for dynamic control based on live brainwave input.
- Architected Imagining Blue — an interactive LED sculpture controlled by a user's brainwaves.
- Invented a modular serial stack enabling 30+ simultaneous serial connections via Arduino.

Pickasound

09/2015 – 01/2017

Lead R&D / Web Audio Developer / Integration Specialist

DSP Architecture Web Audio API Automation

JavaScript / React / Redux / Wavesurfer.js / Web Audio API / DSP.js / Csound / Node.js

- Designed the audio DSP architecture including a standardized effect chain, pitch-shifting, time-stretching, and multi-track recording.
- Created internal and UI/UX unit tests for audio DSP algorithms.
- Developed webpage templates and UI/UX designs.
- Researched and developed fast-rendering audio algorithms.

DSP Concepts

06/2015 – 08/2015

Integration Specialist

Interconnectivity Rapid Prototyping Technical Documentation

Pure Data / AudioWeaver

- Documented legacy code from a previous Pure Data project.
- Developed hardware/software pathways for real-time control of AudioWeaver using Pure Data.

Dave Jones Design

07/2015

Hardware Beta Tester

Eurorack Bug Tracking

Hardware

- Documented hardware performance leading up to the release of the O'Tool+.

Neuropop

01/2015 – 05/2015

Web & Android Developer

Data Visualization Clinical Testing

Android / Data Visualization / Mobile Development

- Designed applications focused on auditory pain solutions for Android.
- Developed auditory patient surveys for clinical testing on Android.
- Maintained and built a web UI workflow for observing and saving patient data.

Qu-Bit Electronics

05/2014 – 11/2014

Audio DSP Developer

Audio DSP Rapid Prototyping QA

Csound / Raspberry Pi

- Created embedded audio DSP algorithms for the Nebulae Eurorack module using Csound.
- Quality assurance and UX/UI testing.

Boulanger Labs

2012 – 2014

Solutions Architect / Integration Specialist

iPad Development HCI Software Architecture

Pure Data / Csound / OpenFrameworks / TouchOSC / Objective-C

- Solutions architect for the Leap Motion app Muse featuring BT — connected OpenGL/OpenFrameworks to Pure Data via OSC, routing gestures into Csound.
- Developed Auto Muse — an algorithmic play-along system using Fibonacci number theory.
- Prototyped audio DSP algorithms and served as UI integration specialist for csSpectral.

Berklee College of Music

06/2013 – 09/2013

ReBoot Tutor

Interactive Performance

Mentorship

MaxMSP / SoftStep

- Assisted professors with interactive development for personal projects.
- Mentored faculty on design paradigms and creative problem-solving.

Berklee College of Music

2010 – 2012

Assistive Music Technology Developer

Music Technology

Data Sonification

HCI

MaxMSP / Arduino

- Invented Optical Sonance — a real-time drawing and sonification machine using Lucas number-based audio to sonify light data as it was drawn.
- Invented wiiAirDrum for the Assistive Music Technology program for Blind and Visually Impaired Musicians at Berklee.
- Software and drawing machines featured at the 2012 President's Advisory Council Meeting hosted by Roger Brown.
- Invented a wireless glove system and interactive robotics project for Nona Hendryx called MIT-1.

EDUCATION

Berklee College of Music

2008 – 2012

Electronic Production & Design / Jazz Composition

Computer Music

Physical Computing

Data Sonification

Sound Design

Jazz Composition

- Studied under Dr. Richard Boulanger creating software and hardware applications using Csound, MaxMSP, Pure Data & C.
- Studied under Tom Zicarelli of Ableton & Cycling74 creating autonomous architectures & creative problem-solving.
- Comissioned to create work for Nona Hendryx createing hardware/software systems for live performances.